Numerical schemes and solvers

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1 Overview

The concentration in the air is advanced by a numerical scheme, chosen as the algorithm passed to `solve`: implicit Euler (`EulerBackward`, the AeoLiS default), `CrankNicolson`, or explicit Euler (`EulerForward`, with the time step limited by the CFL number). Inside a scheme the transport solver is `Trunk` (an upwind discretisation solved as a sparse linear system), `Pieter` (a finite-volume form with fluxes on cell faces), or one of the steady-state solvers, which find the concentration in equilibrium with the current wind at each step: `SteadyState` and `SteadyStatePieter` from AeoLiS, and `SciMLSteadyState`, which poses the equilibrium as a SciML `SteadyStateProblem`. The sparse systems are solved with LinearSolve.jl, so any of its sparse factorizations can be used.

```
ENV["GKSwstype"] = "100"
using Aeolis
using LinearSolve
using Plots
```

2 A test problem

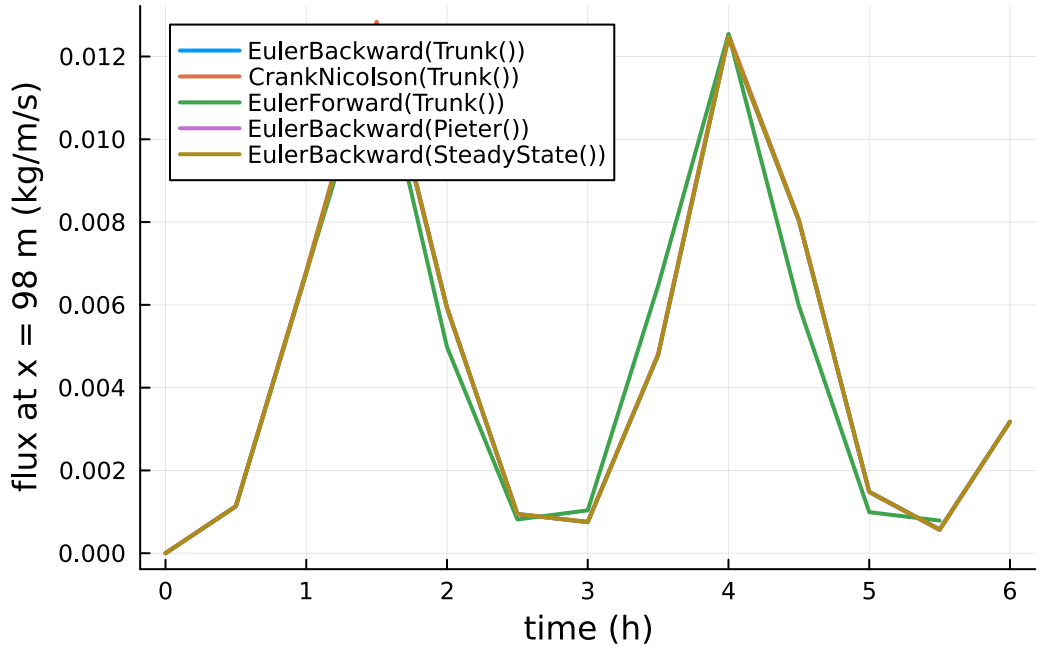
A beach with a varying wind, so that the concentration is not always in equilibrium:

```
x = collect(0.0:2.0:100.0)
t = 0.0:600.0:86400.0
wind = hcat(t, 8 .+ 6 .* sin.(t ./ 3000.0) .^ 2, fill(270.0, length(t)))
p = AeolisParameters(; xgrid_file=x, bed_file=fill(1.0, length(x)), wind_file=wind,
                     grain_size=0.0003, dt=300.0, tstop=6 * 3600.0,
output_times=1800.0,
                     output_vars=[:zb, :qs], CFL=1.0)
prob = AeolisProblem(p)
```

```
AeolisProblem with tspan (0.0, 21600.0)
AeolisParameters
  time: 0.0 to 21600.0 s, dt = 300.0 s
  scheme: euler_backward, solver: trunk
  fractions: 1, layers: 3
  enabled: process_wind, process_transport, process_bedupdate, process_threshold,
  th_grainsize
```

3 Schemes and solvers

```
algs = ["EulerBackward(Trunk())" => EulerBackward(Trunk()),
        "CrankNicolson(Trunk())" => CrankNicolson(Trunk()),
        "EulerForward(Trunk())" => EulerForward(Trunk()),
        "EulerBackward(Pieter())" => EulerBackward(Pieter()),
        "EulerBackward(SteadyState())" => EulerBackward(SteadyState())]
sols = Dict(name => solve(prob, alg) for (name, alg) in algs)
plt = plot(; xlabel="time (h)", ylabel="flux at x = 98 m (kg/m/s)", legend=:topleft)
for (name, _) in algs
    s = sols[name]
    plot!(plt, s.t ./ 3600, transect(s, :qs)[end - 1, 1, :]; lw=2, label=name)
end
plt
```



The explicit scheme takes many more steps, since the CFL number limits its time step to the time the wind takes to cross a cell; the solution statistics count the time steps and the matrix solves:

```
[(name, get(sols[name].stats, :time, 0), get(sols[name].stats, :matrixsolve, 0))
 for (name, _) in algs]
```

```
5-element Vector{Tuple{String, Int64, Int64}}:
 ("EulerBackward(Trunk())", 73, 73)
 ("CrankNicolson(Trunk())", 73, 73)
 ("EulerForward(Trunk())", 116090, 0)
 ("EulerBackward(Pieter())", 73, 72)
 ("EulerBackward(SteadyState())", 73, 0)
```

The bed level after six hours differs by less than a millimetre between the time-dependent schemes; the steady-state solver, which ignores the time the concentration takes to adjust, differs by a few millimetres:

```
ref = transect(sols["EulerBackward(Trunk())"], :zb)[:, end]
[(name, maximum(abs.(transect(sols[name], :zb)[:, end] .- ref))) for (name, _) in
algs]
```

```
5-element Vector{Tuple{String, Float64}}:
 ("EulerBackward(Trunk())", 0.0)
 ("CrankNicolson(Trunk())", 1.137333824896558e-6)
 ("EulerForward(Trunk())", 0.0009280507582671849)
 ("EulerBackward(Pieter())", 1.2569002748152869e-5)
 ("EulerBackward(SteadyState())", 0.005272247071415048)
```

4 Linear solvers

The trunk and pieter solvers assemble one sparse matrix per fraction and time step and solve it repeatedly while the fraction weights are adjusted to the available sediment; the factorization is reused across these solves. A LinearSolve.jl algorithm can be given with `linsolve`:

```
for lin in (nothing, KLUFactorization(), UMFPACKFactorization())
    s = solve(prob, EulerBackward(Trunk(); linsolve=lin))
    d = maximum(abs.(transect(s, :zb)[:, end] .- ref))
    println(rpad(string(something(lin, "default")), 30), " max difference ", d)
end
```

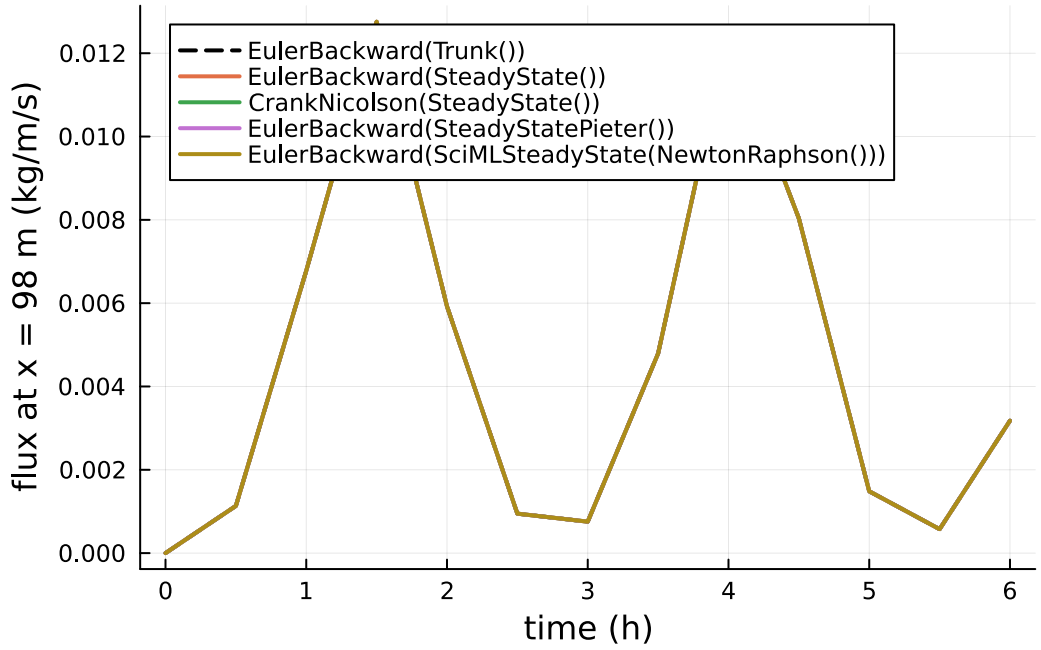
```
default                        max difference 0.0
LinearSolve.KLUFactorization(true, true) max difference 0.0
LinearSolve.UMFPACKFactorization{@NamedTuple{}}(true, true, NamedTuple()) max
difference 0.0
```

5 Steady-state solvers

AeoLiS has three steady-state forms. `SteadyState` sweeps the grid upwind under `EulerBackward` and `EulerForward`, but under `CrankNicolson` it solves the trunk solver's matrix without the time derivative. `SteadyStatePieter` does the same with the pieter solver's face fluxes, under any scheme.

`SciMLSteadyState` is not part of AeoLiS. At each step, and for each fraction, it writes the equilibrium as a `SteadyStateProblem`: the net upwind flux through the cell faces plus a source that relaxes the concentration towards saturation, with erosion over the step limited to the sediment in the top layer. Any SciML steady-state or nonlinear solver can solve it; here `NewtonRaphson` from NonlinearSolve.jl. The source is piecewise linear in the concentration and the problem carries its exact sparse Jacobian, so Newton's method converges in a few iterations.

```
using NonlinearSolve
steady = ["EulerBackward(SteadyState())" => EulerBackward(SteadyState()),
          "CrankNicolson(SteadyState())" => CrankNicolson(SteadyState()),
          "EulerBackward(SteadyStatePieter())" =>
          EulerBackward(SteadyStatePieter()),
          "EulerBackward(SciMLSteadyState(NewtonRaphson()))" =>
          EulerBackward(SciMLSteadyState(NewtonRaphson()))]
ssols = Dict(name => solve(prob, alg) for (name, alg) in steady)
plt = plot(; xlabel="time (h)", ylabel="flux at x = 98 m (kg/m/s)", legend=:topleft)
plot!(plt, sols["EulerBackward(Trunk())"].t ./ 3600,
      transect(sols["EulerBackward(Trunk())"], :qs)[end - 1, 1, :]; lw=2, ls=:dash,
      color=:black, label="EulerBackward(Trunk())")
for (name, _) in steady
    s = ssols[name]
    plot!(plt, s.t ./ 3600, transect(s, :qs)[end - 1, 1, :]; lw=2, label=name)
end
plt
```



The bed level after six hours, against the time-dependent trunk solver:

```
[(name, maximum(abs.(transect(ssols[name], :zb)[:, end] .- ref))) for (name, _) in
steady]
```

```
4-element Vector{Tuple{String, Float64}}:
 ("EulerBackward(SteadyState())", 0.005272247071415048)
 ("CrankNicolson(SteadyState())", 1.8742286556072685e-7)
 ("EulerBackward(SteadyStatePieter())", 1.2569002748152869e-5)
 ("EulerBackward(SciMLSteadyState(NewtonRaphson()))", 0.0007467256756035212)
```

In `SciMLSteadyState` the sediment picked up over a step equals the net flux out of the domain, so the bed change balances transport exactly. AeoLiS's `SteadyStatePieter` balances them only approximately, and with circular lateral boundaries and an alongshore wind it leaves out the flux through one lateral face.

6 References